Untitled1

May 1, 2026

Génération des images 1ére version(Cat edition)

1 Premier essaie de génération d'images

[33]:
```
# Importation des différentes librairies utiles pour le notebook

# Sickit-learn met régulièrement à jour des versions et
# indique des futurs warnings. Ces lignes permettent de ne pas les afficher.
import warnings  # Permet de gérer les avertissements
warnings.filterwarnings("ignore", category=FutureWarning)

# Librairies générales
import pandas as pd  # Manipulation de données en format tabulaire
from scipy.stats import randint  # Pour les distributions aléatoires
import numpy as np  # Calcul numérique
from numpy import mean, std  # Statistiques de base
import os  # Interaction avec le système d'exploitation
from os import listdir  # Liste les fichiers dans un répertoire
from os.path import isfile, join  # Fonctions pour gérer les chemins de fichiers
import sys  # Manipulation du système
import random  # Génération de nombres aléatoires
import cv2  # Librairie pour la vision par ordinateur
import zipfile  # Manipulation des archives ZIP

# Librairie d'affichage
import matplotlib.pyplot as plt  # Visualisation de données
from PIL import Image, ImageDraw  # Manipulation d'images avec Pillow

# Scikit-learn pour l'apprentissage automatique
from sklearn.decomposition import PCA  # Analyse en composantes principales(ACP)

# TensorFlow et Keras pour le deep learning
import tensorflow as tf  # Framework principal pour l'apprentissage profond
import keras  # API de haut niveau pour TensorFlow
from keras import layers  # Modules de construction de modèles Keras
from keras import models  # Définition et manipulation des modèles
from keras import optimizers  # Optimisateurs pour les modèles Keras
```

```
from keras.optimizers import Adam  # Optimiseur Adam
from tensorflow.keras.preprocessing.image import ImageDataGenerator  #Générateur

# Utilitaires Keras pour le traitement d'images
from keras.utils import img_to_array, load_img  # Conversion images en tableaux
from keras.callbacks import (ModelCheckpoint,
                             EarlyStopping)  # Gestion des checkpoints
from keras import backend as K  # Backend Keras pour opérations de bas niveau
from keras.models import Model, load_model  # Définition/chargement des modèles
from keras.layers import LeakyReLU  # Activation LeakyReLU
from keras.preprocessing import image  # Prétraitement des images

# Couches Keras pour les modèles convolutifs et autres
from keras.layers import (Input, Conv2D, Flatten, Dense, Conv2DTranspose,
                          Reshape, Activation,
                          BatchNormalization,
                          Dropout)  # Couches pour CNN et autoencoders
from keras.layers import (InputLayer, MaxPooling2D,
                          UpSampling2D)  # Couches supplémentaires pour les CNN
```

[ ]:
```
# Datasets
from keras.datasets.mnist import load_data
# Scikitlearn
from sklearn.decomposition import PCA # Analyse en composantes principales (ACP)
```

Définition des répertoires pour stocker les modéles appris

[ ]:
```

```

[72]:
```
# Définition du répertoire cible
model_dir = "./Data_humanface_cat_without_caption_1000/"

# Création du répertoire s'il n'existe pas
os.makedirs(model_dir, exist_ok=True)

zip_file = "Data_humanface_cat_without_caption_1000.zip"

!wget https://www.lirmm.fr/~poncelet/Ressources/
 ↪Data_humanface_cat_without_caption_1000.zip


# Extraction du fichier ZIP
with zipfile.ZipFile(zip_file, "r") as zip_ref:
        zip_ref.extractall(model_dir)

# Suppression du fichier ZIP après extraction pour économiser de l'espace
os.remove(zip_file)
```

```
--2026-05-01 23:12:48--  https://www.lirmm.fr/~poncelet/Ressources/Data_humanfac
e_cat_without_caption_1000.zip
Resolving www.lirmm.fr (www.lirmm.fr)... 193.49.104.251
Connecting to www.lirmm.fr (www.lirmm.fr)|193.49.104.251|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 42306490 (40M) [application/zip]
Saving to: ‘Data_humanface_cat_without_caption_1000.zip’

Data_humanface_cat_ 100%[===================>]  40.35M   736KB/s    in 33s

2026-05-01 23:13:22 (1.23 MB/s) - ‘Data_humanface_cat_without_caption_1000.zip’
saved [42306490/42306490]
```

another way to download data idk which is better? Nom local de l’archive et URL source zip$_{f}ile = "Data_{h}umanface_{c}at_{w}ithout_{c}aption{.}zip" zip_{u}rl = "https : //www.lirmm.fr/\ poncelet/Ressources/Data_{h}umanface_{c}at_{w}ithout_{c}aption_{4}00.zip"$

Répertoire cible après extraction src$_{d}ir = "images" image_{d}ir = os.path.join(src_{d}ir, "man")$

Vérifie si le dossier de destination contient des images def humans$_{a}lready_{p}resent() : returnos.path.exists(image_{d}ir)andany(f.endswith(".jpg")forfinos.listdir(image_{d}ir))$

Téléchargement et extraction seulement si besoin if not humans$_{a}lready_{p}resent() : print("AucuneimagetrouvedansCat.Tlchargementetextractionencours...")$

Télécharger si l’archive n’est pas déjà présente if not os.path.exists(zip$_{f}ile) : print(f"Tlchargementdezip_{f}ile...")!wget - Ozip_{f}ilezip_{u}rlelse : print(f"zip_{f}iledjprsent, tlchargementignor.")$

Extraction dans le répertoire courant with zipfile.ZipFile(zip$_{f}ile, "r")aszip_{r}ef : formemberinzip_{r}ef.namelist() : ifnotmember.startswith("_{MACOSX}"):zip_ref.extract(member,".")$

print("Archive extraite dans le répertoire courant.")

Suppression de l’archive après extraction os.remove(zip$_{f}ile)else : print("LedossierHumans/trainexistedjetcontientdesimages.Tlchargementignor.")$

[83]:
```python
# Répertoire contenant les images
src_dir = "Data_humanface_cat_without_caption_1000/images"
image_dir = os.path.join(src_dir, "cat")

files = [f for f in os.listdir(image_dir) if f.endswith(".jpg")]

print("Nombre d'images dans", image_dir, ":", len(files))

if files:
    with Image.open(os.path.join(image_dir, files[0])) as img:
        shape = img.size[::-1] + (3,)  # (height, width, 3)
        print("Format (shape) des images :", shape)
```

```
# Tirage de 5 images aléatoires
random_files = random.sample(files, 5)

# Affichage
plt.figure(figsize=(8, 2))
for i, fname in enumerate(random_files):
    img_path = os.path.join(image_dir, fname)
    img = Image.open(img_path)

    plt.subplot(1, 5, i + 1)
    plt.imshow(img)
    plt.title(fname, fontsize=8)
    plt.axis("off")

plt.tight_layout()
plt.show()
```

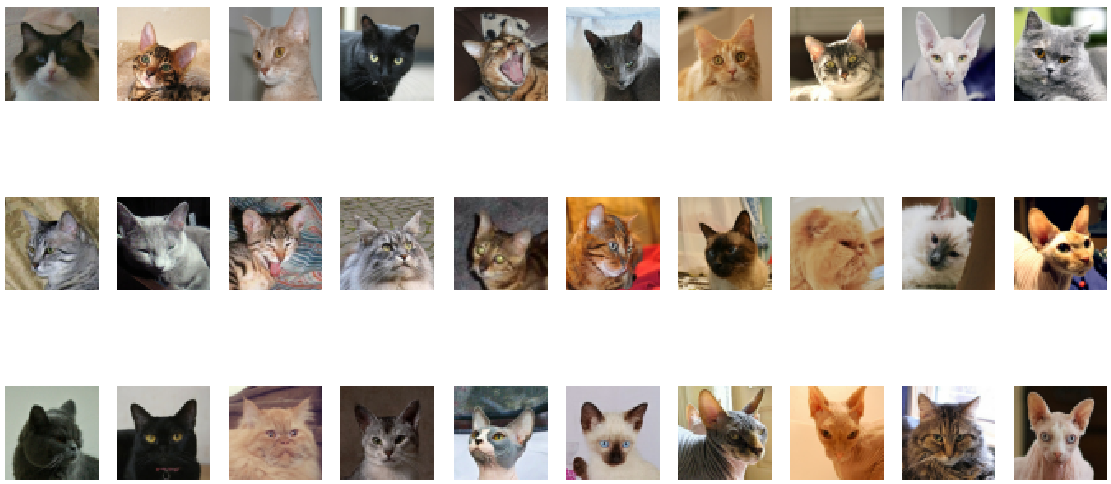
```
Nombre d'images dans Data_humanface_cat_without_caption_1000/images/cat : 1000
Format (shape) des images : (256, 256, 3)
```

2. Importer le jeu de données -

[84]:
```
image_size = (128, 128, 3)  # Dimensions des images (hauteur, largeur, canaux)
batch_size = 512             # Taille des lots
latent_dim = 200             # Dimension du vecteur latent pour l'autoencodeur

# Génération des données normalisées (valeurs entre 0 et 1)
data_flow = ImageDataGenerator(rescale=1./255).flow_from_directory(
    directory=src_dir,
    classes=["cat"],  # Le sous-dossier contenant toutes les images
    target_size=image_size[:2],  # Redimension à 128x128
    batch_size=batch_size,
    shuffle=True,
    seed=42,
    class_mode='input'  # Autoencodeur : entrée = sortie
)
```

```
Found 1000 images belonging to 1 classes.
```

[85]:
```python
# Obtention du premier batch d'images générées
data_batch = next(data_flow)

# Affichage des 30 premières images du batch
import matplotlib.pyplot as plt

fig, axes = plt.subplots(3, 10, figsize=(12, 6))
for i, ax in enumerate(axes.flatten()):
    ax.imshow(data_batch[0][i])
    ax.axis('off')
plt.show()
```



1/ Définition de l'encodeur -

[86]:
```python
def build_encoder(input_dim, output_dim):
    tf.keras.backend.clear_session()

    encoder_input = Input(shape=input_dim, name='encoder_input')
    x = encoder_input

    x = Conv2D(32, kernel_size=3, strides=2, padding='same',
               name='encoder_conv_1')(x)
    x = LeakyReLU(name='encoder_act_1')(x)

    x = Conv2D(64, kernel_size=3, strides=2, padding='same',
               name='encoder_conv_2')(x)
    x = LeakyReLU(name='encoder_act_2')(x)

    x = Conv2D(64, kernel_size=3, strides=2, padding='same',
               name='encoder_conv_3')(x)
    x = LeakyReLU(name='encoder_act_3')(x)

    x = Conv2D(64, kernel_size=3, strides=2, padding='same',
```

```
                name='encoder_conv_4')(x)
    x = LeakyReLU(name='encoder_act_4')(x)

    shape_before_flattening = tf.keras.backend.int_shape(x)[1:]
    x = Flatten()(x)
    encoder_output = Dense(output_dim, name='encoder_output')(x)

    encoder = Model(encoder_input, encoder_output, name='encoder')
    return encoder_input, encoder_output, shape_before_flattening, encoder
```

[87]:
```
# Création de l'encodeur avec les dimensions d'entrée et de sortie spécifiées
encoder_input, encoder_output, shape_before_flattening, encoder = build_encoder(
    input_dim=image_size,
    output_dim=latent_dim
)

# Affichage du résumé de l'encodeur
encoder.summary()
```

```
Model: "encoder"


 Layer (type)                    Output Shape                  Param #

 encoder_input (InputLayer)      (None, 128, 128, 3)                 0

 encoder_conv_1 (Conv2D)         (None, 64, 64, 32)                896

 encoder_act_1 (LeakyReLU)       (None, 64, 64, 32)                  0

 encoder_conv_2 (Conv2D)         (None, 32, 32, 64)             18,496

 encoder_act_2 (LeakyReLU)       (None, 32, 32, 64)                  0

 encoder_conv_3 (Conv2D)         (None, 16, 16, 64)             36,928

 encoder_act_3 (LeakyReLU)       (None, 16, 16, 64)                  0

 encoder_conv_4 (Conv2D)         (None, 8, 8, 64)               36,928

 encoder_act_4 (LeakyReLU)       (None, 8, 8, 64)                    0

 flatten (Flatten)               (None, 4096)                        0

 encoder_output (Dense)          (None, 200)                   819,400
```

```
Total params: 912,648 (3.48 MB)

Trainable params: 912,648 (3.48 MB)

Non-trainable params: 0 (0.00 B)
```

[88]:
```python
def build_decoder(input_dim, shape_before_flattening):
    # Entrée du décodeur (vecteur latent)
    decoder_input = Input(shape=(input_dim,), name='decoder_input')

    # Projeter dans un tenseur de même forme qu'avant le Flatten de l'encodeur
    x = Dense(np.prod(shape_before_flattening))(decoder_input)
    x = Reshape(shape_before_flattening)(x)

    # Couches de déconvolution (effet miroir de l'encodeur)
    x = Conv2DTranspose(64, kernel_size=3, strides=2, padding='same',
                        name='decoder_conv_1')(x)
    x = LeakyReLU(name='decoder_act_1')(x)

    x = Conv2DTranspose(64, kernel_size=3, strides=2, padding='same',
                        name='decoder_conv_2')(x)
    x = LeakyReLU(name='decoder_act_2')(x)

    x = Conv2DTranspose(32, kernel_size=3, strides=2, padding='same',
                        name='decoder_conv_3')(x)
    x = LeakyReLU(name='decoder_act_3')(x)

    x = Conv2DTranspose(3, kernel_size=3, strides=2, padding='same',
                        name='decoder_conv_4')(x)
    decoder_output = Activation('sigmoid', name='decoder_output')(x)

    decoder = Model(decoder_input, decoder_output, name='decoder')
    return decoder_input, decoder_output, decoder
```

Partie décodeur -

[14]:
```python
def generate_from_random_latent(decoder, latent_shape=(7, 7, 8), n=10):
    """
    Tire au hasard n vecteurs latents et les passe dans le décodeur.

    Args:
        decoder: modèle Keras du décodeur.
        latent_shape: forme des vecteurs latents.
        n: nombre d'images à générer.
    """
    # Génération des vecteurs latents
    random_latents = np.random.normal(loc=0.0, scale=1.0,
```

```
                                        size=(n, *latent_shape))
    generated_images = decoder.predict(random_latents)

    # Mise en page cohérente : 5 colonnes par ligne
    n_cols = 5
    n_rows = (n // n_cols) + int(n % n_cols != 0)

    plt.figure(figsize=(n_cols * 4, n_rows * 2))
    for i in range(n):
        ax = plt.subplot(n_rows, n_cols, i + 1)
        plt.imshow(generated_images[i].squeeze(), cmap='gray')
        ax.axis('off')
    plt.suptitle("Images générées à partir de vecteurs latents aléatoires",
                 fontsize=16)
    plt.tight_layout()
    plt.show()
```

[89]:
```
# Création du décodeur en utilisant la dimension de l'espace latent
# et le shape  avant le flattening
decoder_input, decoder_output, decoder = build_decoder(
    input_dim=latent_dim,
    shape_before_flattening=shape_before_flattening
)

decoder.summary()
```

Model: "decoder"

Layer (type)	Output Shape	Param #
decoder_input (InputLayer)	(None, 200)	0
dense (Dense)	(None, 4096)	823,296
reshape (Reshape)	(None, 8, 8, 64)	0
decoder_conv_1 (Conv2DTranspose)	(None, 16, 16, 64)	36,928
decoder_act_1 (LeakyReLU)	(None, 16, 16, 64)	0
decoder_conv_2 (Conv2DTranspose)	(None, 32, 32, 64)	36,928
decoder_act_2 (LeakyReLU)	(None, 32, 32, 64)	0

```
 decoder_conv_3                 (None, 64, 64, 32)             18,464
 (Conv2DTranspose)

 decoder_act_3 (LeakyReLU)      (None, 64, 64, 32)                  0

 decoder_conv_4                 (None, 128, 128, 3)               867
 (Conv2DTranspose)

 decoder_output (Activation)    (None, 128, 128, 3)                 0


 Total params: 916,483 (3.50 MB)

 Trainable params: 916,483 (3.50 MB)

 Non-trainable params: 0 (0.00 B)
```

Assemblage des 2 parties pour former l'autoencodeur -

[90]:
```python
# L'entrée du modèle autoencodeur est celle de l'encodeur
autoencoder_input = encoder_input

# La sortie est obtenue en appliquant le décodeur à la sortie de l'encodeur
autoencoder_output = decoder(encoder_output)

# Création du modèle autoencodeur complet (encodeur + décodeur)
autoencoder = Model(autoencoder_input, autoencoder_output, name='autoencoder')

# Affichage de la structure du modèle
autoencoder.summary()
```

```
Model: "autoencoder"


 Layer (type)                   Output Shape                  Param #

 encoder_input (InputLayer)     (None, 128, 128, 3)                 0

 encoder_conv_1 (Conv2D)        (None, 64, 64, 32)                896

 encoder_act_1 (LeakyReLU)      (None, 64, 64, 32)                  0

 encoder_conv_2 (Conv2D)        (None, 32, 32, 64)             18,496

 encoder_act_2 (LeakyReLU)      (None, 32, 32, 64)                  0
```

```
encoder_conv_3 (Conv2D)          (None, 16, 16, 64)             36,928

encoder_act_3 (LeakyReLU)        (None, 16, 16, 64)                  0

encoder_conv_4 (Conv2D)          (None, 8, 8, 64)               36,928

encoder_act_4 (LeakyReLU)        (None, 8, 8, 64)                    0

flatten (Flatten)                (None, 4096)                        0

encoder_output (Dense)           (None, 200)                   819,400

decoder (Functional)             (None, 128, 128, 3)           916,483


Total params: 1,829,131 (6.98 MB)

Trainable params: 1,829,131 (6.98 MB)

Non-trainable params: 0 (0.00 B)
```

[104]:
```
def train_autoencoder_from_generator(model, train_gen, val_gen,
                                     model_path, epochs=100, patience=5):
    """
    Entraîne un autoencodeur avec des générateurs Keras.

    Args:
        model          : modèle Keras compilé
        train_gen      : générateur d'entraînement
        val_gen        : générateur de validation
        model_path     : chemin de sauvegarde du meilleur modèle
        epochs         : nombre maximal d'époques
        patience       : patience pour EarlyStopping

    Returns:
        Le modèle avec les meilleurs poids restaurés.
    """
    callbacks = [
        EarlyStopping(monitor='val_loss', patience=patience,
                      restore_best_weights=True, verbose=1),
        ModelCheckpoint(filepath=model_path,
                        monitor='val_loss',
                        save_best_only=True,
                        save_weights_only=False,
                        verbose=1)
```

```
    ]

    history = model.fit(
        train_gen,
        validation_data=val_gen,
        epochs=epochs,
        callbacks=callbacks,
        verbose=1
    )

    return model
```

[108]:

```
# Définition des hyperparamètres
epochs = 150
model_name = "catmodeltrain.keras"
model_path = os.path.join(model_dir, model_name)

# Compilation du modèle autoencodeur
autoencoder.compile(
    optimizer='adam',
    loss='binary_crossentropy'
)

# Générateur principal avec normalisation  et séparation train/validation
train_gen = ImageDataGenerator(
    rescale=1./255,
    validation_split=0.2
)

# Générateur pour les données d'entraînement
train_flow = train_gen.flow_from_directory(
    directory=src_dir,
    classes=['cat'],  # le sous-dossier contenant toutes les images
    target_size=image_size[:2],
    batch_size=batch_size,
    shuffle=True,
    seed=42,
    class_mode='input',
    subset='training'
)

# Générateur pour les données de validation
val_flow = train_gen.flow_from_directory(
    directory=src_dir,
    classes=['cat'],
    target_size=image_size[:2],
    batch_size=batch_size,
```

```
    shuffle=False,
    seed=42,
    class_mode='input',
    subset='validation'
)


# Entraînement du modèle autoencodeur
autoencoder = train_autoencoder_from_generator(
    model=autoencoder,
    train_gen=train_flow,
    val_gen=val_flow,
    model_path=model_path,
    epochs=epochs,
    patience=5
)
```

```
Found 800 images belonging to 1 classes.
Found 200 images belonging to 1 classes.
Epoch 1/150
```

```
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
E0000 00:00:1777672681.665284     982 meta_optimizer.cc:967] remapper failed:
INVALID_ARGUMENT: Mutation::Apply error: fanout 'StatefulPartitionedCall/gradien
t_tape/autoencoder_1/encoder_act_1_1/LeakyRelu/LeakyReluGrad' exist for missing
node 'StatefulPartitionedCall/autoencoder_1/encoder_conv_1_1/BiasAdd'.
```

```
2/2  0s 9s/step - loss:
0.6931
Epoch 1: val_loss improved from None to 0.69299, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  22s 13s/step - loss:
0.6931 - val_loss: 0.6930
Epoch 2/150
2/2  0s 1s/step - loss:
0.6929
Epoch 2: val_loss improved from 0.69299 to 0.69282, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  6s 2s/step - loss:
0.6929 - val_loss: 0.6928
Epoch 3/150
2/2  0s 958ms/step - loss:
0.6927
Epoch 3: val_loss improved from 0.69282 to 0.69263, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6927 - val_loss: 0.6926
```

```
Epoch 4/150
2/2  0s 977ms/step - loss:
0.6925
Epoch 4: val_loss improved from 0.69263 to 0.69241, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6924 - val_loss: 0.6924
Epoch 5/150
2/2  0s 1s/step - loss:
0.6922
Epoch 5: val_loss improved from 0.69241 to 0.69215, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6922 - val_loss: 0.6921
Epoch 6/150
2/2  0s 2s/step - loss:
0.6918
Epoch 6: val_loss improved from 0.69215 to 0.69181, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.6918 - val_loss: 0.6918
Epoch 7/150
2/2  0s 1s/step - loss:
0.6914
Epoch 7: val_loss improved from 0.69181 to 0.69134, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.6913 - val_loss: 0.6913
Epoch 8/150
2/2  0s 2s/step - loss:
0.6907
Epoch 8: val_loss improved from 0.69134 to 0.69040, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.6905 - val_loss: 0.6904
Epoch 9/150
2/2  0s 2s/step - loss:
0.6893
Epoch 9: val_loss did not improve from 0.69040
2/2  4s 2s/step - loss:
0.6891 - val_loss: 0.6908
Epoch 10/150
2/2  0s 2s/step - loss:
0.6876
Epoch 10: val_loss improved from 0.69040 to 0.68774, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.6881 - val_loss: 0.6877
```

```
Epoch 11/150
2/2  0s 2s/step - loss:
0.6872
Epoch 11: val_loss improved from 0.68774 to 0.68570, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 3s/step - loss:
0.6866 - val_loss: 0.6857
Epoch 12/150
2/2  0s 1s/step - loss:
0.6841
Epoch 12: val_loss improved from 0.68570 to 0.67402, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6833 - val_loss: 0.6740
Epoch 13/150
2/2  0s 2s/step - loss:
0.6720
Epoch 13: val_loss improved from 0.67402 to 0.66966, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.6711 - val_loss: 0.6697
Epoch 14/150
2/2  0s 2s/step - loss:
0.6695
Epoch 14: val_loss improved from 0.66966 to 0.65946, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.6668 - val_loss: 0.6595
Epoch 15/150
2/2  0s 896ms/step - loss:
0.6605
Epoch 15: val_loss improved from 0.65946 to 0.65592, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6603 - val_loss: 0.6559
Epoch 16/150
2/2  0s 988ms/step - loss:
0.6555
Epoch 16: val_loss improved from 0.65592 to 0.64816, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6549 - val_loss: 0.6482
Epoch 17/150
2/2  0s 2s/step - loss:
0.6498
Epoch 17: val_loss improved from 0.64816 to 0.64309, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
```

```
0.6497 - val_loss: 0.6431
Epoch 18/150
2/2  0s 1s/step - loss:
0.6445
Epoch 18: val_loss improved from 0.64309 to 0.64156, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6439 - val_loss: 0.6416
Epoch 19/150
2/2  0s 2s/step - loss:
0.6412
Epoch 19: val_loss improved from 0.64156 to 0.63635, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.6410 - val_loss: 0.6363
Epoch 20/150
2/2  0s 991ms/step - loss:
0.6376
Epoch 20: val_loss improved from 0.63635 to 0.63247, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6361 - val_loss: 0.6325
Epoch 21/150
2/2  0s 1s/step - loss:
0.6335
Epoch 21: val_loss improved from 0.63247 to 0.62739, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6336 - val_loss: 0.6274
Epoch 22/150
2/2  0s 1s/step - loss:
0.6290
Epoch 22: val_loss improved from 0.62739 to 0.62534, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6285 - val_loss: 0.6253
Epoch 23/150
2/2  0s 2s/step - loss:
0.6259
Epoch 23: val_loss improved from 0.62534 to 0.61949, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.6239 - val_loss: 0.6195
Epoch 24/150
2/2  0s 993ms/step - loss:
0.6208
Epoch 24: val_loss improved from 0.61949 to 0.61505, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
```

```
2/2  5s 2s/step - loss:
0.6203 - val_loss: 0.6150
Epoch 25/150
2/2  0s 2s/step - loss:
0.6166
Epoch 25: val_loss did not improve from 0.61505
2/2  4s 2s/step - loss:
0.6152 - val_loss: 0.6166
Epoch 26/150
2/2  0s 1s/step - loss:
0.6179
Epoch 26: val_loss improved from 0.61505 to 0.60714, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6168 - val_loss: 0.6071
Epoch 27/150
2/2  0s 2s/step - loss:
0.6077
Epoch 27: val_loss did not improve from 0.60714
2/2  4s 2s/step - loss:
0.6088 - val_loss: 0.6071
Epoch 28/150
2/2  0s 863ms/step - loss:
0.6077
Epoch 28: val_loss improved from 0.60714 to 0.59966, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6058 - val_loss: 0.5997
Epoch 29/150
2/2  0s 879ms/step - loss:
0.6023
Epoch 29: val_loss improved from 0.59966 to 0.59592, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.6008 - val_loss: 0.5959
Epoch 30/150
2/2  0s 2s/step - loss:
0.5964
Epoch 30: val_loss improved from 0.59592 to 0.59357, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5965 - val_loss: 0.5936
Epoch 31/150
2/2  0s 2s/step - loss:
0.5916
Epoch 31: val_loss improved from 0.59357 to 0.59175, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 3s/step - loss:
```

```
0.5939 - val_loss: 0.5917
Epoch 32/150
2/2  0s 1s/step - loss:
0.5911
Epoch 32: val_loss improved from 0.59175 to 0.59083, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5924 - val_loss: 0.5908
Epoch 33/150
2/2  0s 963ms/step - loss:
0.5911
Epoch 33: val_loss improved from 0.59083 to 0.59039, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5918 - val_loss: 0.5904
Epoch 34/150
2/2  0s 1s/step - loss:
0.5894
Epoch 34: val_loss improved from 0.59039 to 0.58789, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5901 - val_loss: 0.5879
Epoch 35/150
2/2  0s 3s/step - loss:
0.5898
Epoch 35: val_loss improved from 0.58789 to 0.58634, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 3s/step - loss:
0.5892 - val_loss: 0.5863
Epoch 36/150
2/2  0s 1s/step - loss:
0.5865
Epoch 36: val_loss did not improve from 0.58634
2/2  4s 2s/step - loss:
0.5870 - val_loss: 0.5872
Epoch 37/150
2/2  0s 2s/step - loss:
0.5858
Epoch 37: val_loss improved from 0.58634 to 0.58516, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5876 - val_loss: 0.5852
Epoch 38/150
2/2  0s 2s/step - loss:
0.5843
Epoch 38: val_loss improved from 0.58516 to 0.58475, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
```

```
0.5849 - val_loss: 0.5847
Epoch 39/150
2/2  0s 2s/step - loss:
0.5916
Epoch 39: val_loss improved from 0.58475 to 0.58171, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5871 - val_loss: 0.5817
Epoch 40/150
2/2  0s 2s/step - loss:
0.5812
Epoch 40: val_loss did not improve from 0.58171
2/2  4s 3s/step - loss:
0.5847 - val_loss: 0.5837
Epoch 41/150
2/2  0s 2s/step - loss:
0.5812
Epoch 41: val_loss did not improve from 0.58171
2/2  4s 2s/step - loss:
0.5839 - val_loss: 0.5821
Epoch 42/150
2/2  0s 2s/step - loss:
0.5829
Epoch 42: val_loss improved from 0.58171 to 0.58165, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5819 - val_loss: 0.5817
Epoch 43/150
2/2  0s 1s/step - loss:
0.5800
Epoch 43: val_loss improved from 0.58165 to 0.57966, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5817 - val_loss: 0.5797
Epoch 44/150
2/2  0s 926ms/step - loss:
0.5802
Epoch 44: val_loss improved from 0.57966 to 0.57955, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5804 - val_loss: 0.5795
Epoch 45/150
2/2  0s 916ms/step - loss:
0.5809
Epoch 45: val_loss improved from 0.57955 to 0.57759, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5799 - val_loss: 0.5776
```

```
Epoch 46/150
2/2  0s 2s/step - loss:
0.5788
Epoch 46: val_loss improved from 0.57759 to 0.57726, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5787 - val_loss: 0.5773
Epoch 47/150
2/2  0s 1s/step - loss:
0.5769
Epoch 47: val_loss improved from 0.57726 to 0.57600, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5780 - val_loss: 0.5760
Epoch 48/150
2/2  0s 2s/step - loss:
0.5760
Epoch 48: val_loss improved from 0.57600 to 0.57570, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 4s/step - loss:
0.5771 - val_loss: 0.5757
Epoch 49/150
2/2  0s 1s/step - loss:
0.5767
Epoch 49: val_loss improved from 0.57570 to 0.57454, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5764 - val_loss: 0.5745
Epoch 50/150
2/2  0s 2s/step - loss:
0.5752
Epoch 50: val_loss improved from 0.57454 to 0.57430, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5754 - val_loss: 0.5743
Epoch 51/150
2/2  0s 997ms/step - loss:
0.5737
Epoch 51: val_loss improved from 0.57430 to 0.57353, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5750 - val_loss: 0.5735
Epoch 52/150
2/2  0s 2s/step - loss:
0.5741
Epoch 52: val_loss improved from 0.57353 to 0.57280, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
```

```
0.5742 - val_loss: 0.5728
Epoch 53/150
2/2  0s 927ms/step - loss:
0.5740
Epoch 53: val_loss improved from 0.57280 to 0.57196, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5737 - val_loss: 0.5720
Epoch 54/150
2/2  0s 2s/step - loss:
0.5731
Epoch 54: val_loss improved from 0.57196 to 0.57158, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5731 - val_loss: 0.5716
Epoch 55/150
2/2  0s 1s/step - loss:
0.5734
Epoch 55: val_loss improved from 0.57158 to 0.57109, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5725 - val_loss: 0.5711
Epoch 56/150
2/2  0s 886ms/step - loss:
0.5733
Epoch 56: val_loss improved from 0.57109 to 0.57051, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5720 - val_loss: 0.5705
Epoch 57/150
2/2  0s 2s/step - loss:
0.5692
Epoch 57: val_loss improved from 0.57051 to 0.57007, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5714 - val_loss: 0.5701
Epoch 58/150
2/2  0s 933ms/step - loss:
0.5692
Epoch 58: val_loss improved from 0.57007 to 0.56990, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5710 - val_loss: 0.5699
Epoch 59/150
2/2  0s 1s/step - loss:
0.5712
Epoch 59: val_loss improved from 0.56990 to 0.56928, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
```

```
2/2  4s 2s/step - loss:
0.5707 - val_loss: 0.5693
Epoch 60/150
2/2  0s 1s/step - loss:
0.5696
Epoch 60: val_loss improved from 0.56928 to 0.56904, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5701 - val_loss: 0.5690
Epoch 61/150
2/2  0s 2s/step - loss:
0.5705
Epoch 61: val_loss improved from 0.56904 to 0.56858, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5699 - val_loss: 0.5686
Epoch 62/150
2/2  0s 2s/step - loss:
0.5691
Epoch 62: val_loss improved from 0.56858 to 0.56828, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5693 - val_loss: 0.5683
Epoch 63/150
2/2  0s 2s/step - loss:
0.5674
Epoch 63: val_loss did not improve from 0.56828
2/2  4s 2s/step - loss:
0.5694 - val_loss: 0.5699
Epoch 64/150
2/2  0s 2s/step - loss:
0.5689
Epoch 64: val_loss did not improve from 0.56828
2/2  4s 2s/step - loss:
0.5701 - val_loss: 0.5687
Epoch 65/150
2/2  0s 915ms/step - loss:
0.5670
Epoch 65: val_loss improved from 0.56828 to 0.56746, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5689 - val_loss: 0.5675
Epoch 66/150
2/2  0s 902ms/step - loss:
0.5695
Epoch 66: val_loss improved from 0.56746 to 0.56746, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
```

```
0.5686 - val_loss: 0.5675
Epoch 67/150
2/2  0s 969ms/step - loss:
0.5688
Epoch 67: val_loss improved from 0.56746 to 0.56667, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5681 - val_loss: 0.5667
Epoch 68/150
2/2  0s 2s/step - loss:
0.5682
Epoch 68: val_loss improved from 0.56667 to 0.56664, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5674 - val_loss: 0.5666
Epoch 69/150
2/2  0s 1s/step - loss:
0.5683
Epoch 69: val_loss improved from 0.56664 to 0.56630, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5672 - val_loss: 0.5663
Epoch 70/150
2/2  0s 2s/step - loss:
0.5694
Epoch 70: val_loss did not improve from 0.56630
2/2  4s 3s/step - loss:
0.5676 - val_loss: 0.5663
Epoch 71/150
2/2  0s 1s/step - loss:
0.5672
Epoch 71: val_loss improved from 0.56630 to 0.56568, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5669 - val_loss: 0.5657
Epoch 72/150
2/2  0s 2s/step - loss:
0.5654
Epoch 72: val_loss did not improve from 0.56568
2/2  4s 2s/step - loss:
0.5666 - val_loss: 0.5662
Epoch 73/150
2/2  0s 2s/step - loss:
0.5671
Epoch 73: val_loss improved from 0.56568 to 0.56506, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5665 - val_loss: 0.5651
```

```
Epoch 74/150
2/2  0s 2s/step - loss:
0.5647
Epoch 74: val_loss improved from 0.56506 to 0.56496, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  5s 2s/step - loss:
0.5657 - val_loss: 0.5650
Epoch 75/150
2/2  0s 2s/step - loss:
0.5613
Epoch 75: val_loss improved from 0.56496 to 0.56452, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5656 - val_loss: 0.5645
Epoch 76/150
2/2  0s 2s/step - loss:
0.5633
Epoch 76: val_loss did not improve from 0.56452
2/2  4s 3s/step - loss:
0.5652 - val_loss: 0.5646
Epoch 77/150
2/2  0s 1s/step - loss:
0.5650
Epoch 77: val_loss improved from 0.56452 to 0.56405, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5651 - val_loss: 0.5640
Epoch 78/150
2/2  0s 2s/step - loss:
0.5646
Epoch 78: val_loss improved from 0.56405 to 0.56393, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5647 - val_loss: 0.5639
Epoch 79/150
2/2  0s 2s/step - loss:
0.5640
Epoch 79: val_loss improved from 0.56393 to 0.56358, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5646 - val_loss: 0.5636
Epoch 80/150
2/2  0s 912ms/step - loss:
0.5632
Epoch 80: val_loss improved from 0.56358 to 0.56339, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5642 - val_loss: 0.5634
```

```
Epoch 81/150
2/2  0s 886ms/step - loss:
0.5643
Epoch 81: val_loss did not improve from 0.56339
2/2  4s 1s/step - loss:
0.5640 - val_loss: 0.5635
Epoch 82/150
2/2  0s 965ms/step - loss:
0.5641
Epoch 82: val_loss improved from 0.56339 to 0.56326, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5640 - val_loss: 0.5633
Epoch 83/150
2/2  0s 2s/step - loss:
0.5594
Epoch 83: val_loss did not improve from 0.56326
2/2  4s 2s/step - loss:
0.5639 - val_loss: 0.5634
Epoch 84/150
2/2  0s 2s/step - loss:
0.5626
Epoch 84: val_loss did not improve from 0.56326
2/2  4s 2s/step - loss:
0.5642 - val_loss: 0.5648
Epoch 85/150
2/2  0s 1s/step - loss:
0.5650
Epoch 85: val_loss improved from 0.56326 to 0.56316, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5648 - val_loss: 0.5632
Epoch 86/150
2/2  0s 994ms/step - loss:
0.5629
Epoch 86: val_loss improved from 0.56316 to 0.56274, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5634 - val_loss: 0.5627
Epoch 87/150
2/2  0s 2s/step - loss:
0.5650
Epoch 87: val_loss did not improve from 0.56274
2/2  4s 3s/step - loss:
0.5642 - val_loss: 0.5628
Epoch 88/150
2/2  0s 2s/step - loss:
0.5621
```

```
Epoch 88: val_loss improved from 0.56274 to 0.56245, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5629 - val_loss: 0.5624
Epoch 89/150
2/2  0s 1s/step - loss:
0.5610
Epoch 89: val_loss did not improve from 0.56245
2/2  4s 2s/step - loss:
0.5630 - val_loss: 0.5629
Epoch 90/150
2/2  0s 2s/step - loss:
0.5607
Epoch 90: val_loss improved from 0.56245 to 0.56222, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5625 - val_loss: 0.5622
Epoch 91/150
2/2  0s 1s/step - loss:
0.5646
Epoch 91: val_loss improved from 0.56222 to 0.56182, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5633 - val_loss: 0.5618
Epoch 92/150
2/2  0s 2s/step - loss:
0.5610
Epoch 92: val_loss did not improve from 0.56182
2/2  4s 3s/step - loss:
0.5621 - val_loss: 0.5627
Epoch 93/150
2/2  0s 1s/step - loss:
0.5632
Epoch 93: val_loss improved from 0.56182 to 0.56114, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 2s/step - loss:
0.5627 - val_loss: 0.5611
Epoch 94/150
2/2  0s 983ms/step - loss:
0.5616
Epoch 94: val_loss did not improve from 0.56114
2/2  4s 2s/step - loss:
0.5617 - val_loss: 0.5613
Epoch 95/150
2/2  0s 2s/step - loss:
0.5647
Epoch 95: val_loss improved from 0.56114 to 0.56067, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
```

```
2/2  5s 2s/step - loss:
0.5618 - val_loss: 0.5607
Epoch 96/150
2/2  0s 2s/step - loss:
0.5597
Epoch 96: val_loss did not improve from 0.56067
2/2  4s 2s/step - loss:
0.5614 - val_loss: 0.5614
Epoch 97/150
2/2  0s 2s/step - loss:
0.5638
Epoch 97: val_loss improved from 0.56067 to 0.56051, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5613 - val_loss: 0.5605
Epoch 98/150
2/2  0s 2s/step - loss:
0.5614
Epoch 98: val_loss did not improve from 0.56051
2/2  4s 2s/step - loss:
0.5612 - val_loss: 0.5608
Epoch 99/150
2/2  0s 2s/step - loss:
0.5578
Epoch 99: val_loss improved from 0.56051 to 0.56007, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 3s/step - loss:
0.5610 - val_loss: 0.5601
Epoch 100/150
2/2  0s 2s/step - loss:
0.5630
Epoch 100: val_loss did not improve from 0.56007
2/2  3s 2s/step - loss:
0.5607 - val_loss: 0.5605
Epoch 101/150
2/2  0s 755ms/step - loss:
0.5613
Epoch 101: val_loss did not improve from 0.56007
2/2  3s 1s/step - loss:
0.5606 - val_loss: 0.5601
Epoch 102/150
2/2  0s 1s/step - loss:
0.5579
Epoch 102: val_loss improved from 0.56007 to 0.56002, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5610 - val_loss: 0.5600
Epoch 103/150
```

```
2/2  0s 745ms/step - loss:
0.5614
Epoch 103: val_loss did not improve from 0.56002
2/2  3s 1s/step - loss:
0.5603 - val_loss: 0.5603
Epoch 104/150
2/2  0s 1s/step - loss:
0.5617
Epoch 104: val_loss improved from 0.56002 to 0.55966, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5605 - val_loss: 0.5597
Epoch 105/150
2/2  0s 787ms/step - loss:
0.5604
Epoch 105: val_loss improved from 0.55966 to 0.55948, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 1s/step - loss:
0.5599 - val_loss: 0.5595
Epoch 106/150
2/2  0s 731ms/step - loss:
0.5598
Epoch 106: val_loss improved from 0.55948 to 0.55933, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 1s/step - loss:
0.5600 - val_loss: 0.5593
Epoch 107/150
2/2  0s 1s/step - loss:
0.5612
Epoch 107: val_loss improved from 0.55933 to 0.55898, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5596 - val_loss: 0.5590
Epoch 108/150
2/2  0s 807ms/step - loss:
0.5593
Epoch 108: val_loss did not improve from 0.55898
2/2  4s 1s/step - loss:
0.5593 - val_loss: 0.5592
Epoch 109/150
2/2  0s 808ms/step - loss:
0.5612
Epoch 109: val_loss improved from 0.55898 to 0.55871, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 1s/step - loss:
0.5593 - val_loss: 0.5587
Epoch 110/150
2/2  0s 818ms/step - loss:
```

```
0.5575
Epoch 110: val_loss improved from 0.55871 to 0.55856, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  4s 1s/step - loss:
0.5590 - val_loss: 0.5586
Epoch 111/150
2/2  0s 2s/step - loss:
0.5580
Epoch 111: val_loss improved from 0.55856 to 0.55847, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5588 - val_loss: 0.5585
Epoch 112/150
2/2  0s 1s/step - loss:
0.5601
Epoch 112: val_loss improved from 0.55847 to 0.55832, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5587 - val_loss: 0.5583
Epoch 113/150
2/2  0s 2s/step - loss:
0.5621
Epoch 113: val_loss did not improve from 0.55832
2/2  3s 2s/step - loss:
0.5586 - val_loss: 0.5584
Epoch 114/150
2/2  0s 2s/step - loss:
0.5595
Epoch 114: val_loss did not improve from 0.55832
2/2  3s 2s/step - loss:
0.5588 - val_loss: 0.5585
Epoch 115/150
2/2  0s 1s/step - loss:
0.5573
Epoch 115: val_loss improved from 0.55832 to 0.55819, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5587 - val_loss: 0.5582
Epoch 116/150
2/2  0s 720ms/step - loss:
0.5578
Epoch 116: val_loss improved from 0.55819 to 0.55789, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 1s/step - loss:
0.5583 - val_loss: 0.5579
Epoch 117/150
2/2  0s 1s/step - loss:
0.5568
```

```
Epoch 117: val_loss did not improve from 0.55789
2/2  3s 2s/step - loss:
0.5581 - val_loss: 0.5583
Epoch 118/150
2/2  0s 723ms/step - loss:
0.5599
Epoch 118: val_loss did not improve from 0.55789
2/2  3s 1s/step - loss:
0.5584 - val_loss: 0.5588
Epoch 119/150
2/2  0s 1s/step - loss:
0.5575
Epoch 119: val_loss did not improve from 0.55789
2/2  3s 2s/step - loss:
0.5586 - val_loss: 0.5580
Epoch 120/150
2/2  0s 1s/step - loss:
0.5587
Epoch 120: val_loss improved from 0.55789 to 0.55757, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5578 - val_loss: 0.5576
Epoch 121/150
2/2  0s 1s/step - loss:
0.5608
Epoch 121: val_loss did not improve from 0.55757
2/2  3s 2s/step - loss:
0.5578 - val_loss: 0.5578
Epoch 122/150
2/2  0s 1s/step - loss:
0.5550
Epoch 122: val_loss improved from 0.55757 to 0.55743, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5580 - val_loss: 0.5574
Epoch 123/150
2/2  0s 1s/step - loss:
0.5599
Epoch 123: val_loss improved from 0.55743 to 0.55735, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5574 - val_loss: 0.5573
Epoch 124/150
2/2  0s 1s/step - loss:
0.5561
Epoch 124: val_loss did not improve from 0.55735
2/2  3s 2s/step - loss:
0.5574 - val_loss: 0.5575
```

```
Epoch 125/150
2/2  0s 785ms/step - loss:
0.5561
Epoch 125: val_loss improved from 0.55735 to 0.55720, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 1s/step - loss:
0.5574 - val_loss: 0.5572
Epoch 126/150
2/2  0s 1s/step - loss:
0.5558
Epoch 126: val_loss improved from 0.55720 to 0.55696, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5571 - val_loss: 0.5570
Epoch 127/150
2/2  0s 1s/step - loss:
0.5600
Epoch 127: val_loss improved from 0.55696 to 0.55678, saving model to
./Data_humanface_cat_without_caption_1000/catmodeltrain.keras
2/2  3s 2s/step - loss:
0.5569 - val_loss: 0.5568
Epoch 128/150
2/2  0s 732ms/step - loss:
0.5576
Epoch 128: val_loss did not improve from 0.55678
2/2  3s 1s/step - loss:
0.5568 - val_loss: 0.5568
Epoch 129/150
2/2  0s 729ms/step - loss:
0.5584
Epoch 129: val_loss did not improve from 0.55678
2/2  3s 1s/step - loss:
0.5568 - val_loss: 0.5568
Epoch 130/150
2/2  0s 717ms/step - loss:
0.5579
Epoch 130: val_loss did not improve from 0.55678
2/2  3s 1s/step - loss:
0.5568 - val_loss: 0.5571
Epoch 131/150
2/2  0s 757ms/step - loss:
0.5582
Epoch 131: val_loss did not improve from 0.55678
2/2  3s 1s/step - loss:
0.5573 - val_loss: 0.5579
Epoch 132/150
2/2  0s 684ms/step - loss:
0.5580
```

```
Epoch 132: val_loss did not improve from 0.55678
2/2  3s 1s/step - loss:
0.5579 - val_loss: 0.5569
Epoch 132: early stopping
Restoring model weights from the end of the best epoch: 127.
```

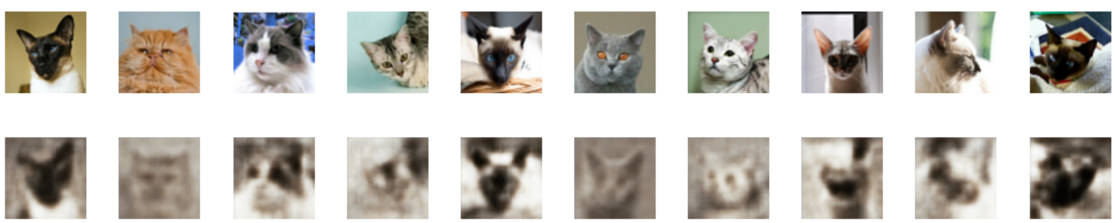
[109]:
```python
def show_images(autoencoder, images=None):
    """
    Fonction pour afficher les images originales et leurs reconstructions␣
↪générées par l'autoencodeur.

    Si aucun lot d'images n'est fourni, la fonction récupère un lot d'images␣
↪depuis le générateur de données.

    Arguments :
    - autoencoder : L'autoencodeur entraîné, utilisé pour générer des␣
↪reconstructions des images.
    - images (optionnel) : Un tableau d'images à afficher. Si non spécifié, un␣
↪lot d'images est récupéré du générateur de données.

    Sortie :
    - Affichage des images originales et leurs reconstructions générées par␣
↪l'autoencodeur.
    """

    if images is None:
        example_batch = next(data_flow)
        example_batch = example_batch[0]
        images = example_batch[:10]

    n_to_show = images.shape[0]

    # Génération des reconstructions des images
    reconst_images = autoencoder.predict(images)

    # Création de la figure pour l'affichage
    fig = plt.figure(figsize=(15, 3))
    fig.subplots_adjust(hspace=0.4, wspace=0.4)

    # Affichage des images originales
    for i in range(n_to_show):
        img = images[i].squeeze()
        sub = fig.add_subplot(2, n_to_show, i+1)
        sub.axis('off')
        sub.imshow(img)

    # Affichage des images reconstruites
```

```
    for i in range(n_to_show):
        img = reconst_images[i].squeeze()
        sub = fig.add_subplot(2, n_to_show, i+n_to_show+1)
        sub.axis('off')
        sub.imshow(img)
```

[110]:

```
# Chemin vers le modèle entraîné
model_path = os.path.join(model_dir, "catmodeltrain.keras")

# Chargement du modèle
autoencoder = load_model(model_path)

# Récupère le premier lot d'images du générateur de données
example_batch = next(data_flow)

# Sélectionne uniquement les images du batch (le premier élément a les images)
example_batch = example_batch[0]

# Prend les 10 premières images du lot pour les afficher
example_images = example_batch[:10]

# Affiche les images originales et reconstruites par l'autoencodeur
show_images(autoencoder, example_images)
```

```
1/1  0s 169ms/step
```



[ ]:

[ ]: